

Shashank S

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EXPERIENCE

Senior Machine Learning Engineer, Walmart

Dec 2024 – Jul 2025

Transitioned to full-time graduate study

- Reduced infrastructure cost growth by **optimizing PySpark jobs** in the adtech ML pipeline: profiled bottlenecks, rewrote shuffle-heavy transformations, and **scaled data processing to match ad volume**.
- Enabled non-technical teams to self-serve analytics by engineering a **Text2SQL agentic system** (plain English queries, visualizations, active chat); accelerated data access and decision-making.
- Improved model reliability and deployment speed by building an **ML observability framework** for production monitoring; standardized ML build, test, and deployment workflows with GitHub Actions.

Machine Learning Engineer III, Glance-InMobi

May 2021 – Nov 2024

Promoted: Data Scientist II to Machine Learning Engineer III, Jul 2022

- Delivered personalized content to **100M DAU** at **sub-50ms p99 latency** by building an end-to-end real-time ML pipeline (prediction service, Vertex feature store, Kafka event streams, ELK logging, Airflow retraining).
- Increased engagement by **5%** and conversion by **12%** by developing and iterating recommendation models (multi-armed bandits, Elo ratings, GBMs, graph-based collaborative filtering) for sparse and dense users.
- Boosted time-spent by **+11.68%** overall and **+18.38%** for cold users by co-authoring lookalike modeling research (SVD + K-Means seeding a MAB model).
- Reduced on-call response time and incidents by owning the ML automation platform end-to-end: streaming pipelines, model controllers, automated rollback triggers, and event-driven personalization.
- Explored diverse ML approaches including user-level *logistic regression*, *reinforcement learning with MABs*, *probability state machines*, *popularity*, and *linear models*.
- Built **automated quiz generation** over live video streams in real-time; PoC demonstrated on YouTube Live Streams.
- Drove A/B testing infrastructure and experimentation culture across the org, increasing experiment velocity and data-driven decision-making.

Data Scientist, Bright Money

Jun 2019 – Apr 2021

Promoted: Junior Research Scientist to Data Scientist, Jan 2021

- Founded the data science function at an early-stage fintech; built the ML layer from scratch across the core product, enabling data-driven features and growth.
- Launched *Debt Manager* on Airflow and Django, driving adoption to **10K+ users** and **\$100K+ daily transaction volume** through end-to-end financial analysis and automated payment scheduling.
- Built *bill identification*, *affordability scoring*, and *overdraft prediction models* from zero-to-one, enabling new product capabilities and risk reduction.
- Developed **forecasting models** for income, expenses, and account balances that powered the product's core retention loop

EDUCATION

Master of Science, Computer Science

May 2026

Georgia Institute of Technology, Atlanta, USA – Specialization: High Performance Computing; GPA 3.87/4.0

- **Research:** Weight Drift Correction in Analog Computing with *Prof. Hyesoon Kim* – paper in preparation (targeting MICRO 2026)
- **Relevant Coursework:** GPU Hardware & Software, Scientific Machine Learning, High Performance Computing, High Perf Parallel Computing, HPC Architecture, Deep Learning, Compiler Design, Designing for Curiosity

Bachelor of Engineering, Computer Science & Engineering

May 2018

Sir M. Visvesvaraya Institute of Technology, Bengaluru, India – First Class with Distinction, 79.8%

- **Thesis:** Image Regeneration with Generative Models
- **Electives:** Pattern Recognition, Clouds & Clusters, Artificial Intelligence

RESEARCH AND PROJECTS

Weight Drift Correction in Analog Computing

- Algorithmic approaches to correcting drift errors in analog MVM crossbar arrays
- Invertible matrix scaling, SVD-based time-invariant decomposition, and RL-based active Correction; evaluated end-to-end on ResNet18/ImageNet under SONOS and PCM Drift models

Project Report: CS 7643

- Explored 3D object reconstruction from 2D images using the Pix2Vox codebase. Developed and evaluated deep learning techniques to infer 3D geometry, automating traditionally manual and costly processes in AR/VR, gaming, medical imaging, autonomous driving, and robotics.

GhostGame.io

- Browser-based online multiplayer word game; users can create and join rooms to play with friends.
- Built on Firebase/GCP with Firestore and Cloud Functions.

Automated Debt Management

- USPTO: [US20220261886A1](#), [US20230075411A1](#)
- R&D contributor to the systems underlying these patents

Deep Learning Segmentation and Quantification of Meibomian Glands

- Published in [Biomedical Signal Processing and Control](#)
- CNN-based segmentation strategy for ophthalmic imaging

Image Regeneration with Generative Models

- An approach to use CapsNet as a discriminator in Generative Adversarial Network and demonstrate its application using semantic inpainting on MNIST and face images — Keras, Tensorflow, Python

Shank's Sessions

- Hosting podcasts and streaming games

Various Tech Toys

- Assorted side projects, mostly built with Streamlit

TECHNICAL SKILLS

Languages: Python, C/C++, PySpark, SQL, Go, Shell, JavaScript, HTML/CSS

Machine Learning & AI: PyTorch, scikit-learn, XGBoost, LightGBM, Recommendation Systems, LLMs, LangChain, Agentic Workflows, Deep Learning, NLP, MLOps, Feature Stores, A/B Testing

High Performance Computing: CUDA, MPI, OpenMP, GPU Profiling, Distributed Systems, Scientific ML (PINNs)

Infrastructure: Spark, Kafka, Docker, Kubernetes, Airflow, Argo, Vertex AI, Prometheus, Grafana, Neptune, Metabase, Django

Cloud & Tools: AWS, GCP, Azure, PostgreSQL, Git, LaTeX, Streamlit, Linux, OpenGL, InkScape, GIMP, Jekyll

HONORS & EXTRAS

ACM ICPC 2015: Qualified to regional levels, honorable mention

Letter of Commendation, Minister of HRD – Highest in Mathematics in Class 12, CBSE Board Exams, India

Section Leader, Code in Place | Stanford University | [2021](#), [2023](#), [2024](#) – Taught Python to 30+ students per cohort

Certifications:

- [Neural Networks & Deep Learning \(deeplearning.ai\)](#)
- [Machine Learning \(Stanford / Coursera\)](#)
- [Android Developer Nanodegree \(Google / Udacity\)](#)